

IGCSE Computer Science CIE

YOUR NOTES



3. Hardware

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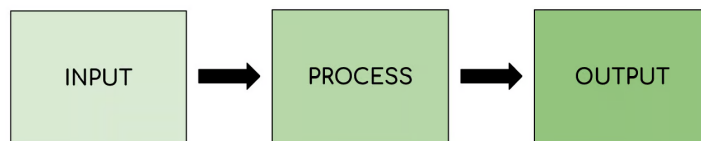
3.1 Computer Architecture

The CPU & Microprocessor

The CPU & Microprocessor

What is the role of the central processing unit (CPU) in a computer?

- A computer system consists of hardware and software
- The main hardware components of a computer system are **input devices**, the **central processing unit**, **primary memory**, **secondary storage** and **output devices**
- Examples of input devices - keyboard, mouse, game controller, sensors, microphone and webcam
- Examples of output devices - monitor, phone screen, speakers, printer, and motors
- Data and commands are inputted by the user using an input device, the central processing unit (CPU) **processes data** by **executing instructions** and the results are outputted to an output device



A diagram showing the input, process, output sequence followed by computer systems

- Below is an example of data being inputted, processed and the results being outputted

Step	Example
Input	• A keyboard is used to input a number
Process	• If the instruction being executed is ADD, the inputted value is added to an existing value
Output	• The result of the calculation is outputted to the user via the monitor

What is a microprocessor?

- A microprocessor is a type of **integrated circuit on a single chip**
- An integrated circuit is a **small electronic circuit** made up of transistors, capacitors, resistors and other electronic components
- The integrated circuit contains a central processor designed to perform **arithmetic and logic operations**, which include adding, subtracting, transferring numbers from one

memory location to another, and comparing two numbers

- The single chip also contains input/output interfaces, and memory
- Microprocessors are a compact way of processing data and can be used in a wide range of electronic devices, including general-purpose computer system and Embedded system

? Worked Example

What is the purpose of a microprocessor?

A		It is the brain of the computer
B		It processes data
C		It executes instructions
D		It is where data is stored permanently

- **BoC [1]**

- The microprocessor is sometimes described as the brain of the computer system but this is not an acceptable exam answer. The purpose of a microprocessor is to process data and execute instructions

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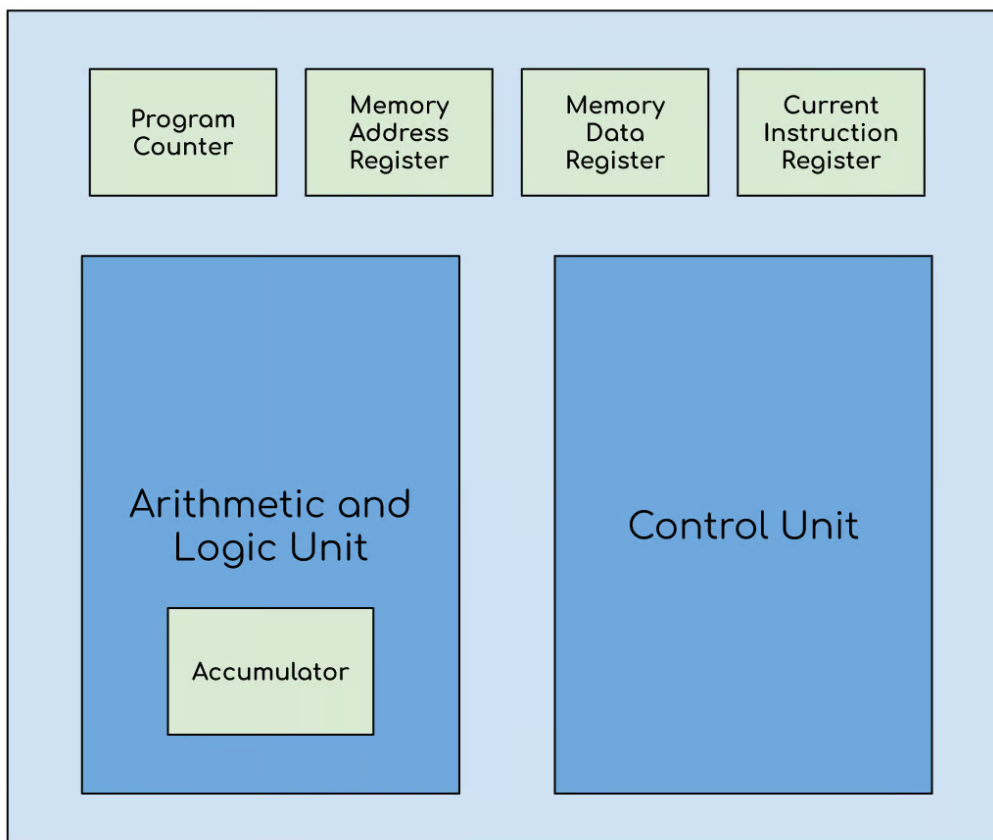
Von Neumann Architecture

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**Von Neumann Architecture**

The components in a CPU, in a computer that has a Von Neumann architecture

- John Von Neumann developed the concept of the **stored program computer** in the 1940s
- The Von Neumann computer architecture which most modern day computers use, is based upon this concept
 - The key feature of the stored program concept, and Von Neumann architecture, is **data and instructions are stored in the same memory (RAM)** as binary
 - Another feature of Von Neumann architecture is a central processing unit (CPU) **fetches instructions** from memory and **executes** them **one at a time (serially)**
 - The CPU then stores the results back into memory

Components of the Central Processing Unit

A diagram showing the main components of the CPU

- The main purpose of the CPU is to **execute instructions** and **process data**
- The CPU has two main components - the **Control Unit (CU)** and the **Arithmetic Logic Unit (ALU)**
 - The Control Unit **controls the flow of data** around the CPU

- The Control Unit also **sends control signals** to the different components instructing them what to do e.g. read, write, add, subtract
- The Control Unit **decodes instructions** (into an opcode and operand)
- The Control Unit controls the **timings of operations** (the clock speed)
- The Arithmetic Logic Unit (ALU) **performs the calculations** required to execute the instructions, these include ADD and SUBTRACT
- The ALU also carries out **logical operations** such as COMPARE
- The ALU has a built-in register where it stores interim results of calculations
- After calculations, the ALU sends data to the MDR
- The CPU also contains a number of registers which are **small memory locations within the CPU**, which temporarily store data needed to execute an instruction
- Special purpose registers have specific roles to play in the execution of an instruction

Special purpose register	Definition
Program Counter (PC)	• stores the address of the next instruction to be fetched from memory
Memory Address Registers (MAR)	• stores the address of the instruction or data to be fetched from or written to memory
Memory Data Register (MDR)	• stores the data that has been fetched from memory or being written to memory • Data from MDR is sent to ALU to be executed
Current Instruction Register (CIR)	• stores the instruction the CPU is currently decoding or executing
Accumulator	• Temporarily stores the results of the calculations performed by the arithmetic and logic unit



Exam Tip

- If asked to describe the purpose of the PC, the MAR or the MDR make sure you explain how the data is being fetched or written to **memory**.

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Worked Example

What are the characteristics of Von Neumann architecture?

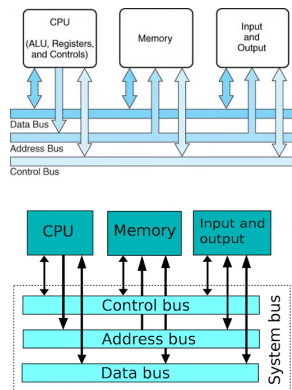
[2]

Any 2 from:

- Both data and instructions are stored in the same memory unit [1]
- Single Arithmetic and Logic Unit [1]
- Single Control Unit [1]
- Uses the **FDE cycle** to execute instructions **serially** [1]

Buses

- Components within the CPU and wider computer system are connected by buses. These are wires down which electronic signals and data travel. The different buses are collectively called **the system bus**
- The system bus is made up of **three different buses** - the **data bus**, the **control bus** and the **address bus**
 - The data bus **transmits data** from the CPU to memory or input/output controllers. It is **bidirectional** which means data can travel in both directions
 - The address bus **transfers addresses from the CPU to memory**. It is **unidirectional** which mean addresses only go **from the CPU to memory**
 - The control bus **transfers control signals** from the control unit to other components in the computer system such as memory or input/output controllers. The control bus is **bidirectional**



http://www.edwardbosworth.com/My5155_Slides/Chapter12/SystemBusFundamentals.htm

A diagram showing how the different buses connect the components in a computer system



Worked Example

Which bus is unidirectional?

[1]

A		Address bus
B		Data bus
C		Control bus
D		System bus

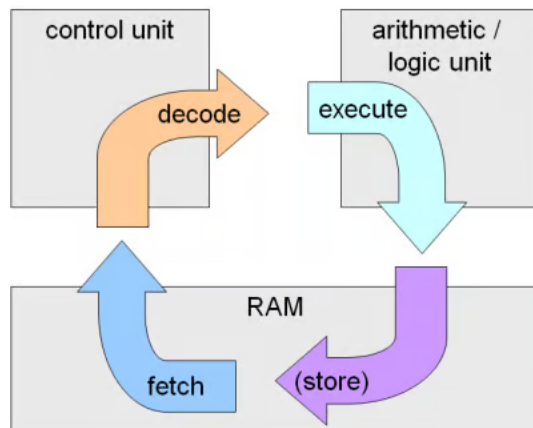
A [1]

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The Fetch, Decode, Execute cycle

- The Central Processing Unit (CPU) executes instructions by performing the Fetch Decode Execute cycle
 - The CPU **fetches an instruction from memory**
 - The **instruction is then decoded** (by the Control Unit) into an **opcode** and an **operand**
 - The **instruction is executed** and the whole cycle is repeated with the next instruction in the process



https://computersciencewiki.org/index.php/The_machine_instruction_cycle

A diagram depicting the Fetch Decode Execute cycle

	Detailed Explanation
Fetch	- The memory address of the instruction to be fetched is stored in the Memory Address Register and is sent down the address bus. - The data/instruction at the memory address is transferred back to the CPU, via the data bus, where it is stored in the Memory Data Register - The instruction is copied into the Current Instruction Register and the Program Counter increments
Decode	The instruction in the Current Instruction Register is decoded, by the Control unit, into an opcode and an operand
Execute	- The instruction is executed by the Arithmetic Logic Unit and the opcode is performed upon the operand. - The result is stored in the accumulator or written to a memory location within memory

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Worked Example

Explain how an instruction is fetched using Von Neumann architecture

[6]

- The Program Counter (PC) holds address/location of the next instruction to be fetched [1]
- The address held in PC is sent to Memory Address Register (MAR) [1]
- The memory address is sent using address bus [1]
- The Program Counter is incremented [1]
- The instruction is sent from the address in memory to the Memory Data Register (MDR) [1]
- The instruction is transferred using the data bus [1]
- The instruction is sent to Current Instruction register (CIR) [1]



Exam Tip

- Make sure you read the question carefully and look at the numbers of marks allocated to judge the level of detail required. Often questions on the fetch-decode-execute cycle only require you to **describe** the steps rather than **explain** how the registers and buses are used during each step (as shown in the table above)

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CPU Performance

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**CPU Performance**

- Central Processing Units can have **multiple cores**
 - A **dual-core** processor has **two cores**
 - A **quad-core** processor has **four cores**
 - Each core runs separate **fetch, decode, execute** cycles, **independently** from one another and **at the same time** (simultaneously) meaning **parallel processing** can take place
 - Multiple cores enables **multitasking** (running more than one program at the same time)
 - Some programs **cannot be split** between cores
 - The **more cores** a computer has the more instructions that can be executed per second resulting in **better performance**
- Each core has a clock speed
 - The clock speed is how many instructions the core can **execute each second**
 - The clock speed is measured in **Hertz**
 - Modern cores can execute billions of instructions per second
 - A **gigahertz** (GHz) is a **billion** instructions per second
 - A megahertz (MHz) is a **million** instructions per second
 - A CPU core with a clock speed of **3.4GHz** can **execute 3.4 billion instructions** per second
- Cache is a **small amount of memory** situated within or close to the CPU with very **fast read/write speeds**
 - It is used for storing **frequently used instructions/data**, recently used instructions, and instructions that are to be fetched and executed next in a process.
 - The impact of increasing the amount of cache is that more data can be stored there and accessed faster **than if it was in RAM** ...which improves the performance of the CPU.
- Double the number of cores does not necessarily mean double the number of instructions executed a second. The cores might have different clock speeds and cache sizes

**Worked Example**

One computer has a single core processor and the other has a dual core processor. Explain why having a dual core processor might improve the performance of the computer

[2]

Any 2 from:

- The computer with the dual core processor has **two cores/double** the amount of cores [1]
- **Parallel processing** can take place [1]
- Each core can execute a separate instruction **at the same time** [1]
- Each core can process instructions **independently** of each other [1]



Exam Tip

- There are three key factors that affect CPU performance - the **number of cores** in your CPU, the **cache size** and **clock speed**. You need to be able to identify these factors and explain how they affect the computer's performance.

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Instruction Sets

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**Instruction Sets**

- An instruction set is a **list of all the commands that can be processed by a CPU**
- Each command has a **unique binary code**
- The table below shows an **example instruction set**. Each instruction has a **mnemonic** that indicates what the instruction does alongside a corresponding binary code

Instruction Set	
ADD	10100001
SUB (subtract)	00100010
LDA (load)	10111111
STR (store)	01100000
BRA (branch)	01011010

- After an instruction is decoded into an opcode and an operand, the CPU finds the opcode in the processor's instruction set. It then knows what operation to perform when executing the instruction

**Worked Example**

Using the instruction set in the table above what would be the operation if the instruction was 00100010 00000010?

[1]

Either of:

- The operation would be SUB [1]
- If the operand was raw data the complete instruction would be to subtract 2 from the value in the accumulator [1]

- Instruction lists are **machine-specific**
 - This means a program created using one computer's instruction set would not run on a computer containing a processor made by a different manufacturer
 - For example, a computer program created using Intel's instruction set would not run on a device containing an ARM processor

Embedded Systems

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Embedded Systems

- An embedded system is a computer systems with a **either one function or limited specific functions** built within a **larger mechanical device**
- Its purpose is to control the device and allow a user to interact with it
- It runs on firmware and does not have additional peripherals
- An embedded system is different to a general purpose computer system like a laptop or desktop computer which can be used to perform many different tasks
- The vast majority of microprocessors manufactured are for use as embedded systems
- Some embedded systems are microcontrollers meaning they are **part of an integrated circuit with built in memory**

Type of embedded system	Description
Microcontrollers	Integrated circuit containing a CPU and memory (RAM or ROM) built in to the same chip
Microprocessor	Integrated circuit containing only a CPU on the chip RAM , ROM , peripherals need to be added

- The microprocessor used in an embedded system is often custom designed
- Embedded systems often have firmware which is software built into the system which cannot be reprogrammed by the user. The software may be able to be updated e.g. GPS software in a car navigation system
- Embedded systems usually have some form of analog or digital input

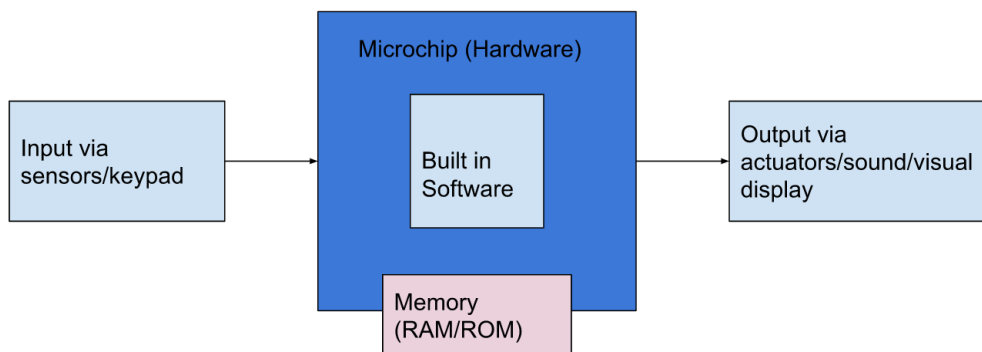


Diagram of an embedded system

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Embedded system	Possible input devices	Possible Output
Digital clock	Buttons to set the time/mode/alarm	Screen showing the time Alarm
Traffic Lights	Pedestrian button Timer Movement sensor	Lights
Lighting system	Movement sensor	Lights
Security system	Keypad to enter alarm code Camera Movement sensor	Alarm
Vending machine	Keypad to make choice	Actuator controlling movement of choice
Central Heating system	Keypad to set temperature Temperature sensor	Heat

- There are many advantages of embedded systems
 - low power consumption
 - small physical size
 - low cost to manufacturer
 - they can be controlled remotely
 - can operate in real time and respond to inputs very quickly



Worked Example

Describe how an embedded system controls a washing machine

[3]

- The user selects the wash cycle they require using a keypad [1]
- The microprocessor will process the inputs and begin to heat the water and move the drum to begin the wash. The actuator controlling the drum and the heating mechanism are the output devices [1]
- Sensors will monitor the water level and temperature [1]

3.2 Input and Output Devices

Input Devices

Input Devices

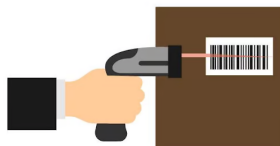
- Every computer system **receives commands** and **data**, from the **real world**, via **input devices**
- Input devices convert inputs into **digital data** which can be processed
- For conventional computer systems, such as desktop computers and laptops, the most common input devices are the **mouse** and the **keyboard**
- There are a wide range of input devices used in more specialist computer systems such as scanners and sensors



Exam Tip

- The exam board has limited the number of devices you may be asked about to eight - keyboard, optical mouse, microphone, barcode scanner, digital camera, QR code scanner, touch screen (resistive, capacitive and infra-red), 2D and 3D scanners
- If you're asked to list some input devices, check you're not listing one included in the question as you won't get a mark for this
- You won't be asked how they technically work but may be asked what each device does, why it does it and when it may be used

Barcode Scanner



CREATED BY VECTORPORTAL.COM

<https://vectorportal.com/vector/scanning-a-barcode-/21323>

Diagram of a barcode being scanned by a handheld barcode scanner

- Barcodes are a series of black and white bars which **represent a code**
- A barcode reader shines a **red laser** at the barcode to read the data it represents
- The light from the white lines is reflected back
- Black lines absorb the light so less is reflected
- The different levels of reflection are **converted into a binary value** which can be processed by a microprocessor
- A barcode reader can be handheld or built into a larger machine such as a self-service checkout at a supermarket

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- Barcodes can be used in many ways such as **identifying a product** being sold or **tracking a package** through the delivery process
- Benefits include **faster checkouts**, **automated stock control**, **less chance of error** due to manual entry of data, and more **detailed tracking information**

? Worked Example

How are barcodes used in supermarkets to manage stock control?

When a customer purchases an item, its barcode is scanned in at the checkout [1]

The barcode reader sends the digital code for the product to the stock database where it is removed from the stock inventory [1]

This means the number of items of stock is always accurate and up to date [1] and the supermarket staff can reorder items quickly when stock is low [1]

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Digital Camera

- A digital camera works by **capturing light** and converting it into a digital image
- Light enters the camera through the lens, it reaches an image sensor where it is split into millions of pixels (small squares). Each pixel measures light intensity which is **converted into binary** and represents a colour.
- Digital cameras are **integrated into smartphones**, used in **security systems** and by **professional photographers** to create high quality digital images
- An advantage of digital cameras is they show a **preview** of the image
- They also **instantly create an image** which can then be easily **duplicated** and **transmitted** via bluetooth or WiFi
- Software can be used to **edit digital photos**, for example applying a filter or retouching a photo

Keyboard

- A keyboard is the most common device used for **text-based data input**
- They are connected either by a USB cable or wirelessly to the computer system
- They are **built into laptops**.
- **Smartphones** and tablets have **virtual keyboards**
- Each key on a keyboard has a peg underneath it which makes contact with a conducting membrane. This is then converted into an electrical signal to transmit a **unique character code**

Microphone

- A computer microphone works by **converting sound waves into electrical signals** that can be processed by the computer.
- They can capture **any real world sound** and **convert it into digital data** which can be stored, duplicated or modified
- The microphone has a diaphragm that vibrates in response to sound waves. These vibrations are then converted into electrical signals by a coil of wire attached to the back of the diaphragm. Changes in the signal are recorded by a microprocessor using an analogue to digital converter

- Microphones are used to **record music, telephone calls, communicate online** and **dictation**

Optical Mouse

- A **mouse** **use a laser to detect and track movement**, this is then processed by a microprocessor which interprets the movement and replicates it when **moving a virtual cursor on-screen**
- Items can be **selected or moved** using the left mouse button, whilst the right button usually displays additional menus
- They can be **wired or wireless**
- They are used to control the cursor in a Graphical User Interface (GUIs)
- They are simple to use and provide the user with an **intuitive** way to navigate the computer
- They are **reliable** due to no moving parts

QR code scanner



<https://pixabay.com/vectors/qr-code-scanner-bar-code-156717/>

Diagram of a QR code being scanned by a mobile device

- A QR code is a **visual representation of data using black and white squares**
- QR codes can represent **over 7000 digits** whereas a barcode represents up to 30 digits
- QR codes are **scanned by a camera** (often on a mobile device)
- A piece of software convert the squares into **binary data**
- QR codes often **link to a website** where more information can be found
- They can also be used to **advertise products, share contact details, provide promotional codes, train tickets, and event tickets**

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Worked Example

Describe how festival organisers could make use of QR codes and QR code scanners

[4]

Any 4 of:

Festival e-tickets could be in the form of a QR code [1]

QR scanners could scan them on entry to the festival [1]

QR codes could be used to advertise the festival [1]

QR codes could be used to link to the festival website [1]

QR codes could be used to provide information during the festival [1]

Touchscreen

- A touchscreen can be classed as both an input device and an output device

Types of touch screen

Type	How it works	Benefits/Drawbacks	Uses
Resistive screen	Resistive screens consist of two conductive layers. The top layer is flexible. When the screen is touched the two layer connect, completing a circuit	+ Cheap to produce. + Resistant to surface contaminants + Can be activated with nearly every object (stylus, finger, gloved hand)	Used in cash machines, information kiosks, medical equipment
Infra-red screens	LEDs shine infrared light across a screen forming a matrix. When the screen is touched the beams are interrupted	+ Excellent image quality, + high precision + durable + allows for multiple touches at the same time - Requires a bare finger or stylus for activation	Tablets, laptops, smartphones



Capacitive screens	Made up of a protective layer, a transparent conductive layer and a glass substrate. Touching the screen changes the electrostatic field of the conductive layer	+Excellent image quality +Unlimited touch-life +Scale well - Sensitive to interference from light, water, snow	Large scale commercial displays Information kiosks Medical equipment
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2D and 3D scanners

- A two-dimensional (or flat-bed) scanner shines a strip of light over a document. Reflected light is measured for each pixel. This data is converted into binary data so the document can be **digitally recreated**
- A three-dimensional laser scanner shines a light over an object. The **geometry and dimensions** are recorded so the **object can be recreated digitally**
- The digital 3D model can be **modified using specialist software**
- A 3D printer could then print out the model
- 2D scanners can be used for creating **digital versions of documents or photographs**
- Reading passports** at airports
- 3D scanners can be used to create 3D models for use with **computer-aided design (CAD)** software
- Creating **replicas** with 3D printers
- Typical uses of 3D scanners and printers are dentistry, product development, medical
- A cheap and quick way of creating a **digital representation** of both 2D and 3D objects so they can be **manipulated, stored, transmitted or copied**

Summary of input device uses and benefits

Device	Example Uses	Benefits
Barcode scanner	Checkouts, parcel delivery	Faster checkout, less errors, detailed stock/tracking information
QR code scanner	Advertising products, linking to websites, e-tickets	Can hold more information than a barcode Can be scanned using a camera on a mobile device (smartphone)
2D and 3D scanners	Scan documents, photos and objects	Converts 2D and 3D objects into digital data which is easier and quicker to transmit/duplicate or manipulate

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Digital camera	Smartphones, professional photography, security systems	Instant preview and image
Keyboard	Entering text	Universal, easy to use
Mouse	Navigating a GUI Making selections and entering commands	Intuitive
Microphone	Recording real-world sound Musical recordings Telephone calls Dictation	
Touchscreen	Entering text and commands on mobile devices (tablet, smartphone) Cash machines Interactive advertisements Information screens	Can be durable Good image quality Intuitive/easy to use



Exam Tip

- The exam board will refer to a touchscreen as an input device.

Output Devices

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Output Devices

- An output device shows the **results of the processing** in a way humans can understand
- In a general purpose computer system, the main output device is the **monitor/screen**
- Other output devices include **projectors, printers, speakers and actuators**



Exam Tip

- If you're asked to list some output devices, check you're not listing one included in the question as you won't get a mark for this

Actuators

- **Actuators** are used in conjunction with a motor to translate energy (electrical, air, hydraulic) into **real-world movement of a physical object**.
- They come in a **variety of sizes** and have been made specifically for a particular function.
- Examples of use include **turning a wheel, opening or closing a door, controlling a conveyer belt, operating machinery, moving robotic arms, vibrating a machine, starting or stopping a pump, opening or closing a valve**
- They are often used with **sensors**. The input of the sensor is checked against **stored values**. If the input is within a certain range an actuator is used to provide movement of a physical object



Worked Example

A theme park has a game where a player has to run from the start to the finish without getting wet. The system for the game uses sensors and a microprocessor to spray water at a player as they run past each sensor

Describe how the sensors and the microprocessor are used in this system

[6]

- Motion sensors are used to detect the player's movement [1]
- The sensor sends data to microprocessor [1]
- Data is compared with stored data [1]
- if value is within range water is sprayed [1]
- signal sent to actuator to spray water [1]
- if value is outside of range no action is taken [1]

Light projectors

- Used to project computer outputs onto a **large screen**
- Used to give presentations in both **business** and **education** settings
- There are two types - **Digital light processing** (DLP) projectors and **Liquid crystal display** (LCD) projectors.
- **DLP** systems use millions of micro-mirrors arranged in a grid on a microprocessor within the projector. Light is shone through **colour filters and the mirrors**. The position of the mirrors



can be altered to change the intensity of the light

- **LCD** projectors use **three mirror filters** to separate an image into red, green and blue wavelengths. The three images are then combined to produce the full colour image which is passed through the lens on to the wall/screen

Device	Advantages	Disadvantages
Digital light projector (DLP)	Higher contrast ratios. Smooth video	Image tends to suffer from "shadows" when showing a moving image
	Higher reliability	DLP do not have grey components in the image
	Quieter running than LCD projector.	The colour definition is not as good as LCD projectors
	Smaller and lighter	
	Better suited to dusty atmospheres than LCD projectors	
LCD projector	Give a sharper image than DLP projectors	Contrast ratios are not as good as DLPs
	Better colour saturation and intensity than DLP projectors	Have a limited life
	Use less power and generate less heat	LCD panels degrade over time.

Laser and Inkjet printers

- Both printers create **hard copies** of a digital document
- Hard copies are needed when you do not have access to an electronic device

Inkjet printers:

- Have a print head which moves across the page
- **Spray liquid ink droplets** from ink cartridges through very fine nozzles onto the paper (these are in the print head)
- The droplets can be **piezoelectric** or **thermal bubble** technology
- Inkjet printers produce **high quality hard copies of digital images** or documents

Laser printers:

- **Are very fast** when making **multiple copies** of a document
- Are useful for **high volume** print jobs for example producing **leaflets**
- Have a **low running cost per page**.
- Are often used in **business** and education setting
- Have **large toner (dry ink) cartridges** and large paper trays

- Use positive and negative charges to make the toner particles stick to piece of paper, heat is then used to bond the particles to the paper

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Worked Example

Match the characteristic to the correct type of printer

Characteristic	Inkjet	Laser
Uses ink cartridges		
Uses toner (powdered ink)		
Great for bright colours and images		
Great for clear, sharp images and text		
Widely used in homes and home offices with lower print volumes		
Widely used in office with higher print volumes		

[1 mark per row]

Characteristic	Inkjet	Laser
Uses ink cartridges	x	
Uses toner (powdered ink)		x
Great for bright colours and images	x	
Great for clear, sharp images and text		x
Widely used in homes and home offices with lower print volumes	x	
Widely used in office with higher print volumes		x

3D printers

- 3D printers create **3D models of a digital model** by **depositing layers of material**, such as plastic or resin, on top of one another to slowly build up a 3D object
- They allow for **precision** and can be used in medicine to create prosthetics and blood vessels

- 3D printers can be used to create **rapid prototypes**
- Models can be **transmitted digitally** and then models printed out all across the world

Liquid Crystal Display screens

- LCD screens are made up of **millions of tiny liquid crystals**.
- The display is made of **pixels** arranged in a **matrix**
- The display requires backlighting
- They are used for **TVs, monitors, tablets** and **phones**
- They have **low power consumption** and run at a **cool temperature**
- They **do not suffer image burn** or flicker issues
- They **provide bright images** and colours
- They are **cheaper to produce** than LED screens



Worked Example

Describe three benefits of LCD screens

[3]

Any 3 of:

- Low power consumption [1]
- Runs at a cool temperature [1]
- Bright image/colours [1]
- Do not suffer Image burn [1]
- Cheaper to purchase than an LED screen [1]

Light emitting diode (LED) screen

- An LED screen is made up of **tiny LEDs**
- They do **not need backlighting** which makes them **thinner** and **lighter** to hang on the wall
- LED screens are replacing LCD displays in TVs, monitors, laptops and mobile devices
- They have **better image quality** and a **longer life span**
- They can be used to **create very large screens** that can be used to show visual content at events such as football matches and music festivals
- They consume **very little power** making them suitable for being switched on for many hours in the day

Organic light emitting diode (OLED) screen

- **Much thinner and lighter** than traditional LCD screens
- Use **organic light emitting diodes** (OLEDs).
- They use **organic carbon compound** to create semiconductors
- No form of back lighting is required
- You can have **very thin, flexible** screens

Speakers

- Speakers are used to take digital sounds or recordings and output them as sound waves which can be heard by humans

YOUR NOTES



- The digital data is changed into a electric current using a **digital to analogue converter** (DAC)
- It is then passed through an **amplifier** to create a current large enough to drive a loudspeaker
- The loudspeaker converts the current into a sound wave
- Typical uses include **listening to music**, listening to **video sound**, **telephone calls** and **alarms**

? Worked Example

What would an appropriate output device be for a large screen showing replays during a football match?
Justify your choice.

[3]

- An LED screen [1]
- Any 2 of:
 - Consume little power [1]
 - Can be used to create large screens [1]
 - Provide a better quality image compared to alternatives [1]

💡 Exam Tip

- This section of the course has been simplified for exams taking place from 2023 onwards. In the past students needed to know how each device physically worked. You now only need to know:
 - What each device does and why it does it
 - When it is used
- Older textbooks may contain more detail than is required

YOUR NOTES



Sensors

YOUR NOTES



Sensors

- Sensors are **input devices**
- They **measure a physical property of their environment** such as light levels, temperature or movement
- Sensors can be used for monitoring systems and control systems
 - A monitoring system **tracks the state of a system**, it gathers data and may issue **warning messages**
 - A control system will **control the system** based upon the input from sensors
 - For example, if the water temperature in a fish tank fell below the acceptable level, a control system would **start up a heater**. The system will then continue to check the water temperature readings and when they are within the acceptable range it will **switch off the heater**
 - This process of the outputs affecting the following set of input values received from the sensors is called a feedback loop



Exam Tip

- You are likely to be given a scenario and asked to identify the sensors that would be used.
- The 14 sensors you need to be aware of are in the table below

Sensor type	What it measures	Typical use
Acoustic	Sound levels	To detect changes in sound levels of industrial machinery To monitor noise pollution In security system to detect suspicious sounds
Accelerometer	Acceleration rate, tilt, vibration	Detecting sudden changes in vehicle movement and deploy safety features if needed In mobile phones to detect orientation of the device
Flow	Rate of gas, liquid or powder flow	Detect changes in the flow through pipes in water system

Gas	Presence of a gas e.g. carbon monoxide	Detect levels of gas in confined spaces Detect gas levels when fixing gas leaks
Humidity	Levels of water vapour	Monitor humidity in greenhouses
Infra-red	Detecting motion or a heat source	Security systems detecting intruders who break the beam Measures heat radiation of objects - used by emergency services to detect people
Level	Liquid levels	Detects levels of petrol in a car tank Detect levels of water in a water tank Detect a drop in water levels due to a leak
Light	Light levels	Automatically switching on lights when it gets dark (street lights, headlights)
Magnetic field	Presence and strength	Anti-lock braking system Monitoring rotating machinery such as turbines
Moisture	Presence and levels of moisture	Monitoring moisture in soil Monitoring damp in buildings
pH	Acidity or alkaline	Monitoring soil to ensure optimum growing conditions Monitor pH levels in chemical processes

YOUR NOTES





Pressure	Gas, liquid or physical pressure	Monitoring tyre pressure Monitoring pressure in pipes during manufacturing process
Proximity	Distance	Monitoring position of objects in robotics Used in safety systems to prevent objects colliding
Temperature	Temperature in Celsius, Fahrenheit or Kelvin	Used to maintain temperature in swimming pools Used to control temperature in chemical processes



Worked Example

What type of sensor would be used to help drivers when reversing?

[1]

A		Motion
B		Infra-red
C		Light
D		Proximity

- **D [1]**
- A proximity sensor would be used to monitor how close the car was to objects. It would then sound a warning alarm when the proximity is below a certain value (meaning the car is too close to an object)

Is this a monitoring system or a control system?

[1]

A		Monitoring
B		Control

- **A [1]**
- The reversing warning system is a monitoring system. It indicates to the user when the car is too close to an object but it does not stop the car automatically

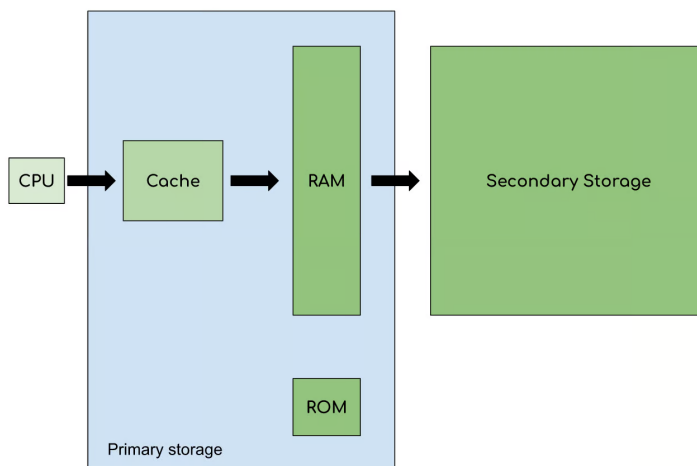


3.3 Data Storage

Primary Storage

Primary Storage

- Primary storage is **directly accessible by the CPU**, it includes **RAM, Cache, and ROM**
- The purpose of Random Access Memory (RAM) is to **store data, instructions and software (including parts of the OS) currently in use**
 - RAM is also called **main memory**
 - RAM has **faster read/write speeds** than secondary storage
 - RAM is **volatile** which means it is **temporary** memory and **all data will be lost** when the **computer is switched off**
 - The **larger the main memory**, the more data the computer access quickly, which can **improve the computer's performance**
 - A standard desktop computer has between 4GB and 16GB of RAM
- The parts of the operating system currently in use will be stored in RAM
- The purpose of cache is to store **frequently/recently used instructions and data**
 - Cache has **faster reads/write speeds than RAM**
 - There are different levels of cache
 - Level 1 cache is situated within the CPU
 - Level 2 and 3 cache are situated on the motherboard
 - If you **increase the amount of cache** you can store more frequently used data and **increase the performance** of your computer
 - If you have too much cache it will decrease the data access speeds because the data requested will take longer to find
 - Cache is **volatile** and **all data will be lost** when the computer is **switched off**



A diagram showing the parts of a computer system which are classed as primary storage

- The role of Read-only memory (ROM) is to store the computer's **boot-up sequence**
 - The boot-up sequence is a set of instructions the computer executes when the computer is **initially switched on**

- This sequence will run a series of checks to ensure all of the system **hardware is working**
- It will then **load the operating system** which will move from secondary storage to RAM
- ROM is **non-volatile** which means data will be stored when the computer is switched off
- ROM is **read-only** which means data cannot be written to ROM

The difference between ROM and RAM

RAM	ROM
Volatile	Non-volatile
Can read and write data to RAM	Read-only



Exam Tip

- Make sure you're specific about which component you're referring to in exam questions and give the proper name e.g. RAM rather than memory unit



Worked Example

Describe what is meant by main memory and how it is used in the Von Neumann model for a computer system

[3]

Any 3 of:

- Memory is RAM [1]
- It is primary storage [1]
- It is volatile memory [1]
- It holds currently in use data/instructions [1]
- It is directly accessed by the CPU [1]

YOUR NOTES



Secondary Storage

YOUR NOTES



Secondary Storage

- Secondary storage is **not directly accessed by the CPU**
- It is used for **permanent** storage of data
- Secondary storage is **non-volatile**. This means data is stored even when power is lost to the device
- Secondary storage is needed as primary storage (such as RAM) is volatile and will not retain data when the computer is turned off
- Unlike ROM, secondary storage **can be read from and written to**. ROM is read only
- Common examples of secondary storage include **hard disk drives (HDDs)**, **solid-state drives (SSDs)**, **external hard drives**, **USB flash drives**, **CDs**, **DVDs**, and **Blu-ray discs**. These can be used to store data to transfer it to another computer
- Data access speeds for secondary storage devices are typically slower than primary storage (such as RAM)
- Secondary storage devices often have **large capacity** (can store large amounts of data)
- **Software** such as the operating system is saved **permanently on secondary storage**. The instructions and data that makes up the software then moves to RAM when it is in use
- Secondary storage devices can be internal (inside the computer) and external (outside the computer)

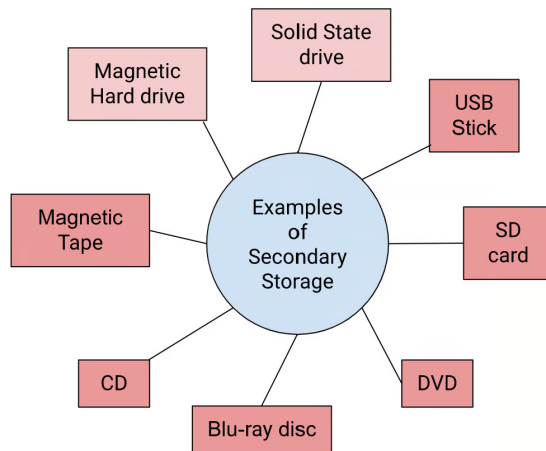


Diagram showing different internal and external secondary storage devices

Comparison of primary and secondary storage

Primary storage	Secondary storage
Volatile / temporary	Non-volatile / permanent
RAM and cache can be read from and written to ROM is read-only	Secondary storage can be read from and written to

YOUR NOTES



Can be directly accessed by the CPU	Cannot be directly accessed by the CPU
Smaller capacity	Larger capacity
Data access speeds faster than secondary storage	Data access speeds slower than primary storage
Stores data and instructions	Stores data and instructions



Worked Example

Which of the following is an example of secondary storage?

A	Random access memory
B	Read only memory
C	Solid State Drive
D	Blu-ray disc

- **C and D [1]**
- Random access memory and read only memory are both primary storage and can be accessed directly by the CPU. Solid-state drives and blu-ray discs are classed as secondary storage. Data can be permanently stored on them but they cannot be accessed directly by the CPU



Exam Tip

- When referring to a USB memory stick, don't refer to it as USB as this could be anything that is plugged in with USB and not necessarily a storage device



Types of secondary storage

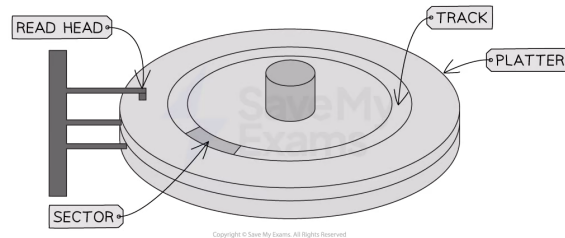
- There are three types of secondary storage – **magnetic**, **solid state** and **optical**

Exam Tip

- You need to be able describe the operation of each type of secondary storage. You may also need to recommend a type of storage for a given scenario and justify your choice

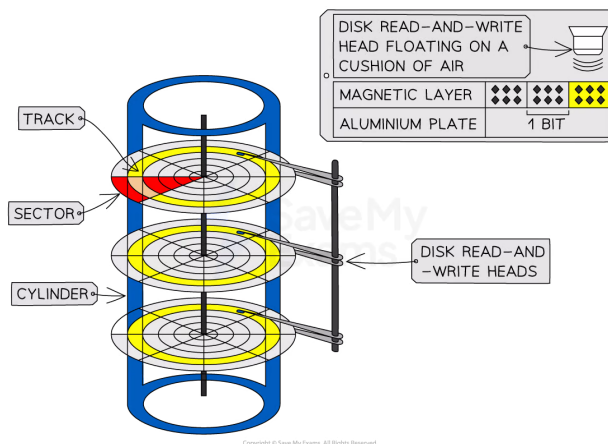
Magnetic

- Magnetic hard disk drives have been the main type of internal secondary storage in personal computers for many years however solid state drives are increasingly popular due to their durability, low power consumption and high read/write speeds
- A magnetic hard disk is made up of **several metal discs** coated with a magnetic material
 - These are called **platters**
 - Iron particles on each platter are **magnetised to represent a 0 or 1**



A diagram showing the structure of a magnetic hard disk

- Each platter is divided by concentric circles creating several **tracks** and wedge shaped **sectors**. Where they intersect is a **track sector**



A diagram showing how each platter is divided by tracks and sectors

- The hard drive **spins the metal disk(s)** at a high speed (typically around 5400–7200 RPM) using a **motor**

- A **read/write arm**, controlled by an actuator, **moves the head** over the surface of the disc to the location of the data
- The data is read/written using **electromagnets**

Solid State

- Solid State secondary storage is made of transistors that are arranged in a grid layout
- It uses **NAND** and **NOR** gates in electrical circuits to persistently control the flow of electrons
- Current flowing is 1, no current is 0 - which is how electrical flow can represent binary values
- A NAND gate will take in two inputs and produce an output (1/electrical current) as long as both inputs are not 1

The truth table for a NAND gate

InputA	InputB	Output
0	0	1
0	1	1
1	0	1
1	1	0

- A NOR gate will produce an output (1) only when both inputs are 0

The truth table for a NOR gate

InputA	InputB	Output
0	0	1
0	1	0
1	0	0
1	1	0

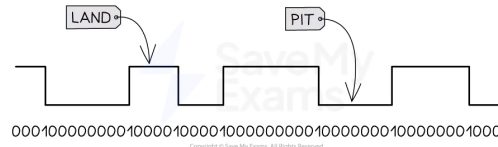
Optical

- Optical devices include **CDs**, **DVDs** and **Blu-rays**
 - Blu-rays have the largest capacity
 - CDs have the lowest capacity
 - CD-R are **read-only** (you cannot save data on to them)
 - CD-RW can be **written to and read from**
 - **DVD-RW** can be **written to and read from**

YOUR NOTES



- All optical devices work by shining a laser at the disk and **processing the reflection**
- An arm moves the laser across the surface of the disk
- In **CD-Rs a laser burns the data, permanently on to the disk**, by creating **pits** and **lands**
- The laser is also used to read the data from the pits and lands
- When the laser light hits the point where the pit changes into a land or vice versa the light scatters and is not reflected back as well. This is captured by a sensor and can be interpreted as a change in the binary value



A diagram showing how data is burnt on to optical discs to create lands and pits which can then be used

- In CD-RW the chemical composition of the disk is changed to represent 0s and 1s. This change can be overwritten



Exam Tip

- Make sure you have a greater understanding of the operation of the different types of storage



Worked Example

Describe how a magnetic storage device stores data

[6]

Any 6 of:

- Storage device has platters [1]
- Platters/disk divided into tracks [1]
- Storage platter/disk is spun [1]
- Has a read/write arm that moves across storage media [1]
- Read/writes data using electromagnets [1]
- Uses magnetic fields to control magnetic dots of data [1]
- Magnetic field determines binary value [1]

YOUR NOTES



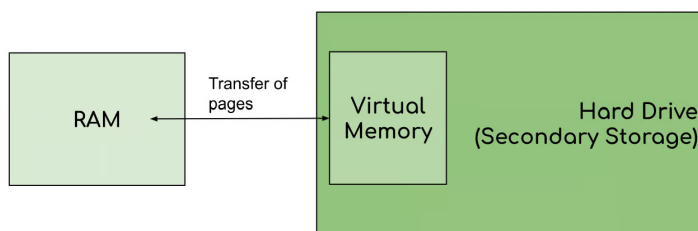
Virtual Memory

YOUR NOTES



Virtual Memory

- The purpose of RAM is to store current data and program instructions in use
- The amount of physical RAM available is limited. The average modern desktop computer has between 4GB and 16GB of RAM. Some **programs are larger** than this
- Virtual memory is part of **secondary storage** which **acts as RAM when RAM is full**
- It **prevents** the computer system **crashing** when **RAM is full**
- Data that needs to be accessed by the CPU directly is switched by the operating system from virtual memory into RAM and an inactive page from RAM replaces it in virtual memory. This is called **page switching**



A diagram showing page switching between RAM and virtual memory

- The table describes the steps that are taken when a program is opened on a computer

Step	Description
1	• The memory manager software built into the Operating System checks to see if there is sufficient space in RAM to move the new program's instructions into RAM, so that they can be directly accessed by the CPU
2	• If RAM is full, the memory manager will move out the most inactive pages of data and put them in virtual memory until needed
3	• The new program instructions are moved into RAM



Worked Example

Sandy is playing an open world role playing computer game set in a large 3D environment. This requires the use of virtual memory. Explain why

[3]

- To extend the RAM capacity [1]
- To stop the computer game from crashing when the physical RAM is full [1]
- To allow the computer to process the large amount of data required [1]

YOUR NOTES

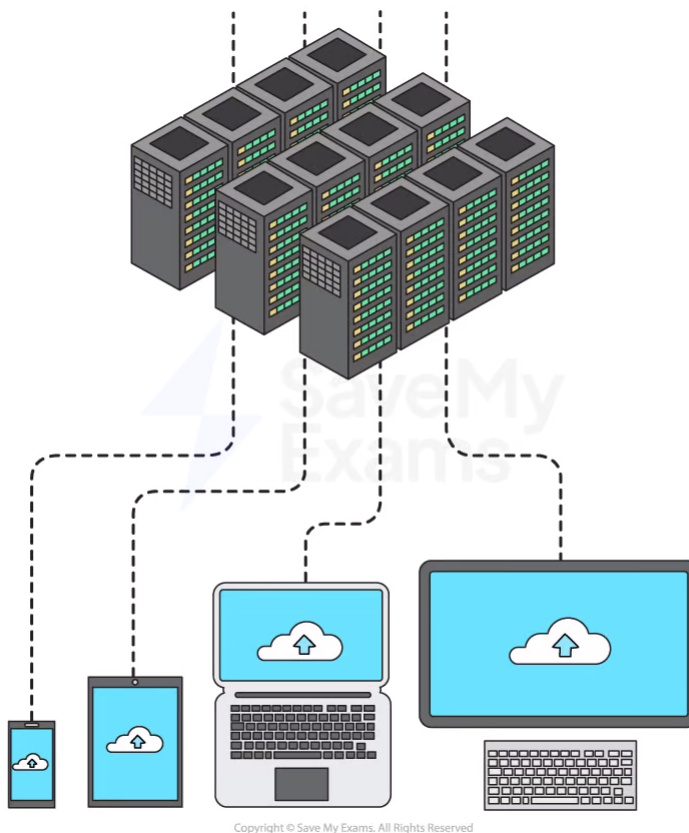


Cloud Storage

YOUR NOTES

**Cloud Storage**

- Cloud storage is the concept of storing data on remote servers, that can be accessed via the internet, instead of on a local storage device (such as an SSD or HDD)
- The physical servers, where the data is stored, are owned and managed by a hosting company
 - Cloud storage warehouses often have thousands of servers containing harddrives
 - The hosting company provides services such backing up the data, managing access to the data as well as general security
- The data can be accessed at any time, on any device, as long as there is an internet connection
- The data is accessible using a web browser or dedicated software that connects to the cloud storage service
- There is potentially limitless storage, although it does need to be paid for
- Online storage means it is easy to share data and collaborate with others
- If the internet connection is slow/lost the access to your data is affected



A diagram showing different devices connecting to remote servers

Advantages of cloud storage

- Cloud storage offers several advantages over storing data on a local secondary storage device

Benefit	Explanation
Accessibility	• Data can be accessed from anywhere • Data can be accessed by anyone with the relevant permissions, making it quick to share files and collaborate with others • Data can be accessed on any device with an internet connection
Scalability	• Cloud storage providers allow customers to increase or decrease their storage capacity as needed
Reliability	• Cloud storage providers often use multiple servers to store and backup data, reducing the risk of data loss due to hardware failure
Security	• Cloud storage providers offer advanced security features, such as data encryption and multi-factor authentication multi-factor authentication, to protect user data from unauthorised access • There is no need to hire specialist staff as IT services being provided by the cloud storage provider

- You also do not need to purchase local secondary storage devices
- One cloud storage centre is more **environmentally friendly** than millions of individual servers
- There are some drawbacks to using cloud storage
 - If the user has a slow or **unreliable internet connection** they would have problems **accessing** their files
 - If you are storing a lot of data it can be **expensive**
 - You have to pay the cloud storage provider
 - You may need to commit to an annual subscription
 - You may have to pay for high download and upload data transfer limits
 - Data being sent over the internet has the potential to be **intercepted**
 - Security is managed by the cloud storage provider which means the user **does not have complete control** over it but is still legally responsible for ensuring any personal and sensitive data is secure

YOUR NOTES





Worked Example

A school currently stores all of their data on local servers. They are considering upgrading their system to utilise cloud storage.

Explain two disadvantages to the school of storing their data in the cloud.

[4]

- You need a constant internet connection [1]
- if the internet connection failed, teachers would not be able to access lesson files [1]
- if the internet connection failed, teachers would not be able to access student data/take electronic registers [1]
- if the internet connection failed, students would not be able to access their work/coursework [1]
- Reliant on third party to carry out security procedures [1]
- but the school is still legally responsible if data is accessed by unauthorised people [1]
- Data stored on the cloud could be hacked or intercepted [1]

Explain two advantages to the school of storing their data in the cloud.

[4]

- Additional storage can be purchased if needed [1]
- The cloud hosting company provides back up [1]
- The cloud hosting company provides security [1]
- Data can be accessed from anywhere and from any device provided there is an internet connection [1]
- So teachers would be able to access data from home [1]
- So students would be able to access work from home [1]

3.4 Network Hardware

NIC

Network Interface Card

- A Network Interface Card (NIC) or Network Interface Controller is an **internal physical component** which enables computers and other devices to **connect to a network**
- They are also known as **network adapters**
- NICs have a built in ethernet port and can be **connected to a network via an Ethernet cable**
- You can have Wireless Network Interface Cards (WNIC) which allow a device to wirelessly connect to a network
- The primary function of a NIC is to send and receive **data packets** between the computer or device and the network
- The **NIC converts data to be sent into signals that can be sent across the transmission medium**
 - If this is through a wired network, the signals will be voltages through an ethernet wire or pulses of light through a fibre
 - If this is through a wireless network, the signals will be radio waves
- NICs are likely to be integrated into the motherboard but can be added as an expansion card

YOUR NOTES



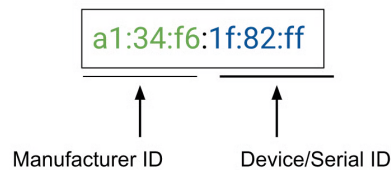
MAC address

YOUR NOTES



MAC Addresses

- A Media Access Control/MAC address is used to identify a device on a network
- It is a **unique code** which consists of **six pairs of hexadecimal codes** separated by a colon
- Each pair of hex digits is a Byte of data. So in total a MAC address is **6 bytes** or 48 bits long
- An example MAC address is **b4:71:ac:f3:21:a2**
- The MAC address is **assigned to the NIC** by its **manufacturer**
- The first **three pairs are the manufacturer ID** number and the last three pairs is the serial number of the NIC and **identifies the device** on the network



An example MAC address – the first part identifying the manufacturer and the second part identify the device

- A MAC address cannot be changed. It is **static**

A table listing the differences between an IP address and a MAC address

MAC Address	IP Address
Permanently assigned to a NIC and cannot be changed	Can be used to identify devices on a network but are dynamic and can be changed
MAC addresses are written in hexadecimal	IP addresses are written in denary
MAC addresses are 6 bytes long	IP addresses are 4 bytes long
MAC addresses are used to identify a device on a LAN	MAC addresses are used to identify a device on a WAN



Worked Example

Describe what is meant by a MAC address

[4]

Any 4 of:

- Used to identify a device [1]
- It is a unique address [1]
- It is a static address/it does not change [1]
- It is set by the manufacturer [1]
- The first part is the manufacturer ID [1]
- The second part is the serial number of the device [1]

YOUR NOTES



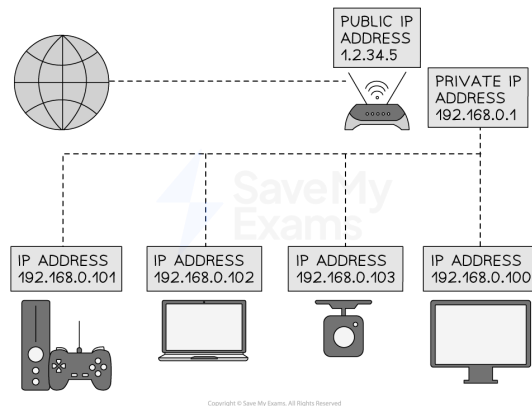
IP Address

YOUR NOTES



IP Address

- An IP (Internet Protocol) address is a **unique numerical identifier** assigned to devices connected to the internet
 - IP addresses are used to identify and **locate devices** on the internet
 - This **allows data** to be sent and received between devices
- An IPv4 address is **four** numbers separated by a **full stop**
 - Each number is a byte and range between **0–255**
 - There are over 4 billion unique addresses using the IPv4 system
 - For example, an IPv4 address might look like this: 192.164.3.1
- Routers and gateways have a **unique public address**
- Public addresses** are assigned by your internet service provider and is the IP address for the network
- Public addresses are often static addresses meaning they do not change



A diagram showing how Private and Public IP addresses are used to identify devices on the Internet and within a LAN

- A router **can assign private IP addresses** to devices connected within its network
- These are often **dynamic IP addresses** meaning they can **change**
- Dynamic addresses are **assigned** from a list of available addresses **at the time they are required**
- The IP4 system is running out of possible addresses due to the enormous increase in networked devices
- A new system called **IPv6** has been developed. It provides enough unique addresses for every networked device on the planet
 - IPv6 uses a **128 bit address**
 - Digits are organised into **eight** groups
 - Each group is made up of four hexadecimal digits
 - For example bb43:ff3f:0000:549b:e43e:db84:1f1f:0b9e



Worked Example

Give two characteristics of an IP address

[2]

Any 2 of:

- A unique address [1]
- Can be public or private [1]
- Can be static or dynamic [1]
- IPv4 has four groups of digits [1]
- Each number is between 0–255 [1]
- and separated by a full stop [1]
- IPv6 has eight groups of digits [1]

A comparison of IPv4 and IPv6 IP addresses

IPv4	IPv6
32-bit address	128 bit address
Four groups of denary digits separated by a full stop	Eight groups of four hexadecimal digits separated by a colon
Over 4.3 billion unique addresses	Almost infinite amount of unique addresses



Worked Example

Explain the difference between a private and a public IP address

[4]

Any 4 of:

- A private IP address is not accessible from the internet [1]
- and is used to identify a device within a local area network [1]
- A public IP address is used to identify a device on the internet [1]
- Public IP addresses are assigned by an internet service provider (ISP) [1]
- Public IP addresses are unique [1]
- Public IP addresses can be accessed from anywhere on the internet [1]
- Public IP addresses allow devices to communicate with each other across different networks [1]

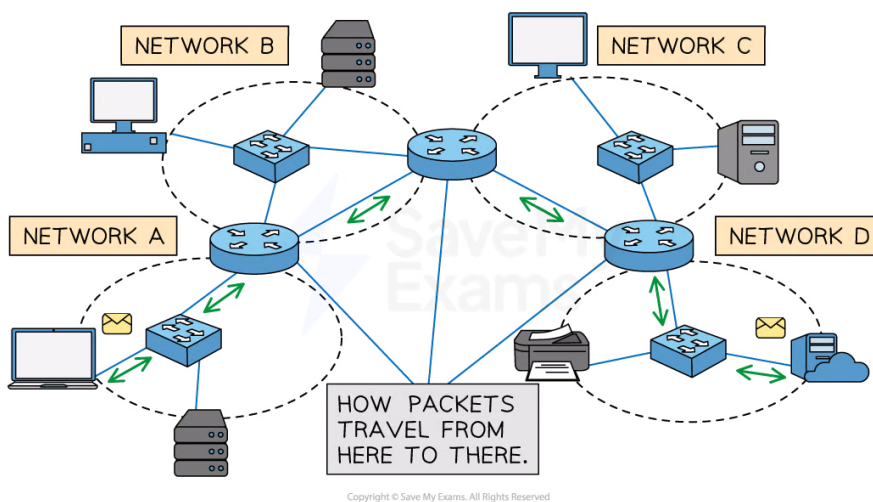
Router

YOUR NOTES



Router

- A router is an important piece of network hardware. It connects **one network to another network**
 - This means it can be used to **connect a local area network (LAN) to the internet**, which is a wide area network(WAN)
- An important role of the router is **to analyse data packet** and **direct** them on to their destination



A diagram showing multiple networks connected by routers, represented by the blue circular objects

- A router being used to connect a LAN to a WAN will have a **public IP address**, which has been assigned to it by an Internet Service Provider
- It is this public IP address which other routers use to identify and direct packets to the network
- An important role of the router is **to analyse data packets** and **direct** them on to their destination
 - The **IP address of both the sender and intended recipient** is stored in the **header** of the data packet
 - The **header** contains **information about the packet**
 - The **payload** is the **actual data** being sent
- If the data packet is coming into the LAN, the router will send the data packet to the **specific device**, within its LAN, that the packet is meant for
- If the packet is being sent from a device within the LAN, it will read the header of the packet to determine the intended **destination IP address**
 - It will then **forward the packet** on to the destination
 - It might have to **travel through several routers** before it gets to its destination

- Each pass from router to router is called a **hop**
- The network access device or 'home hub' used in your home network will have a router built into it

A summary of the steps a router takes when it receives a data packet

Step	Description
1	• a router receives incoming data packets from one network and analyses the packet header to determine the destination IP address
2	• It then looks up the IP address in a routing table (routing table of known networks) to determine the next network where the packet should be sent
3	• The router then forwards the packet to the appropriate network or device

- This process is repeated by every router the data packet passes through, until it reaches its destination
- In addition to routing data between networks, routers can also perform other functions such as...
 - **assigning IP addresses** to devices within the LAN
 - **filtering incoming traffic** based on certain criteria, such as IP address, port number, or protocol type

YOUR NOTES



